



Intersectional Disparities in ML: A 29-Group Fairness Analysis of UCI Adult Income Prediction

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Abstract—This paper presents a comprehensive intersectional fairness audit of machine learning–based income prediction using the UCI Adult dataset. Traditional single-attribute fairness assessments—focused solely on gender, race, or age independently—fail to capture compounded disadvantages experienced by individuals situated at the intersection of multiple demographic identities. To address this limitation, we evaluate fairness across 29 intersectional subgroups derived from Gender $\times$ Race $\times$ Age combinations, using Statistical Parity Difference (SPD), Disparate Impact Ratio (DIR), and Equal Opportunity Difference (EOD) relative to the privileged Male_White_Middle-aged reference group. Our analysis reveals severe disparities: the most disadvantaged groups exhibit DIR values as low as 0.04 and SPD values near -0.11 , falling well below the legal four-fifths (0.8) disparate-impact threshold. We further show that these extreme inequities are completely masked under standard single-attribute fairness analysis, which conceals the compounding effects of intersecting identities. These findings underscore the necessity of intersectional audits as a prerequisite for deploying ML models in high-stakes socio-economic contexts.

Index Terms—algorithmic fairness, disparate impact, equal opportunity, intersectional bias, machine learning fairness, UCI Adult dataset

I. INTRODUCTION

MACHINE learning (ML) systems increasingly inform decisions in high-stakes domains such as employment screening, credit risk assessment, healthcare prioritization, and criminal justice. These models often operate at national or institutional scale, shaping access to economic opportunity, financial stability, and essential services. As their influence continues to expand, regulatory bodies, researchers, and practitioners have expressed growing concern about the potential for ML systems to perpetuate—or even amplify—historical patterns of discrimination embedded within the data on which they are trained. Ensuring that automated decision-making pipelines function fairly for all demographic groups has thus become a critical requirement, especially in contexts where biased outcomes can have profound and long-lasting societal effects.

Fairness research in machine learning has traditionally focused on evaluating disparities along **single demographic attributes**, such as gender *or* race *or* age. Widely used metrics such as **Statistical Parity Difference (SPD)**, **Disparate Impact Ratio (DIR)**, and **Equal Opportunity Difference (EOD)** typically compare model outcomes between two groups defined along a single dimension—for example, male vs. female or White vs. non-White. While these attribute-specific evaluations are valuable, they are fundamentally limited: they implicitly assume that disadvantages operate independently across demographic categories. In practice, however, many forms of discrimination emerge from the interaction of multiple social identities, meaning that individuals who belong to more than one marginalized group may experience disproportionately worse outcomes than would be predicted by any single attribute alone.

This insight forms the foundation of **intersectionality**, a conceptual framework introduced by Crenshaw to describe how overlapping social identities can create unique forms of disadvantage. In real-world applications, intersectionality manifests in numerous ways: for instance, an employer’s automated screening system may appear equitable when comparing men vs. women or White vs. Black applicants individually, yet may yield systematically poorer outcomes for specific subgroups such as **Black women under age 30**. These intersection-specific harms remain invisible under traditional single-attribute fairness assessments, even though they may reflect some of the most severe and actionable forms of bias. Such cases illustrate why intersectional fairness analysis is not merely a methodological refinement, but a practical necessity for evaluating ML systems deployed in consequential socio-economic settings.

Despite growing recognition of intersectional harms, **current literature contains comparatively few systematic, large-scale intersectional fairness audits of widely studied ML benchmarks**. Most existing work evaluates fairness along one attribute at a time or considers only a small number of intersectional combinations. As a result, important disparities—particularly those affecting small or highly specific demographic groups—remain understudied. Furthermore, the UCI Adult Income dataset, one of the most extensively used fairness benchmarks in ML

research and education, has rarely been examined through a rigorous intersectional lens. This gap limits our understanding of how algorithmic disparities emerge and propagate when demographic categories interact.

This paper addresses this research gap by presenting the **first comprehensive 29-group intersectional fairness analysis of ML-based income prediction on the UCI Adult dataset**. We define intersectional subgroups across the combination of **Gender $\times$ Race $\times$ Age**, yielding 29 distinct demographic groups. For each subgroup, we compute standard fairness metrics (SPD, DIR, EOD) using the commonly assumed **Male_White_Middle-aged** group as the privileged baseline. The resulting analysis uncovers significant and previously unreported disparities: the most disadvantaged groups exhibit **DIR values as low as 0.04** and **SPD values near -0.11** , magnitudes that fall *far below* the widely recognized **0.8 four-fifths rule** used in legal disparate impact assessment. These values reflect severe inequities that would have remained undetected under conventional single-attribute evaluation.

A further contribution of this work is a direct comparison between **single-attribute fairness metrics** and **intersectional fairness metrics**. Our results provide empirical evidence that evaluating fairness along only one attribute at a time **masks the most extreme disparities**, giving a false sense of equity. For example, gender-based analysis may suggest relatively balanced outcomes between men and women overall, while race-based analysis shows modest differences between White and non-White groups. Yet, intersectional analysis reveals that specific subgroups—such as *Female_Black_Young*—experience disproportionately worse outcomes, even though this disadvantage is not visible in any single-axis assessment. This discrepancy highlights the critical limitations of current fairness evaluation pipelines and reinforces the necessity of multi-dimensional auditing for models deployed in sensitive socio-economic contexts.

The remainder of this paper is organized as follows. **Section III** reviews related work on fairness metrics, bias in ML systems, and prior efforts in intersectional analysis. **Section IV** describes the dataset, subgroup construction procedure, fairness metrics, and analytical methodology. **Section V** presents the empirical results, including subgroup-level disparities, violations of the four-fifths rule, and comparisons between single-attribute and intersectional evaluations. **Section VI** concludes the paper with implications for future fairness auditing, limitations of the present work, and directions for mitigating intersectional bias in ML systems.

II. RELATED WORK

Research on algorithmic fairness spans several interconnected areas, including classical fairness metrics, single-attribute bias analyses, intersectionality theory, emerging intersectional ML research, and prior fairness

evaluations on the UCI Adult dataset. This section surveys these domains and motivates the need for a comprehensive 29-group intersectional analysis.

A. Classical Fairness Metrics

A foundational component of fairness research involves defining quantitative measures that compare model behavior across demographic groups. Three of the most widely used metrics in machine learning are **Statistical Parity Difference (SPD)**, **Disparate Impact Ratio (DIR)**, and **Equal Opportunity Difference (EOD)**.

Statistical Parity Difference (SPD) measures the difference in favorable prediction rates between a protected group and a privileged group. SPD stems from early social science work on discrimination detection and was later formalized in algorithmic fairness frameworks such as the disparate impact removal method proposed by Feldman et al. [X]. Negative SPD values indicate disproportionately lower positive predictions for the protected group.

Disparate Impact Ratio (DIR) is defined as the ratio of favorable outcome rates between protected and privileged groups. DIR is closely tied to U.S. legal standards such as the **EEOC’s four-fifths rule** [X], which states that a DIR below 0.8 may signal unlawful disparate impact. This legal foundation has made DIR especially influential for auditing ML systems in employment and credit contexts.

Equal Opportunity Difference (EOD) focuses on differences in true positive rates, ensuring that qualified individuals across demographic groups receive similar benefits from the model. Hardt et al. [X] formalized EOD as part of their broader definition of “equality of opportunity” in supervised learning.

Together, these metrics form the core of contemporary fairness assessments but are traditionally applied along **single demographic dimensions**.

B. Single-Attribute Fairness Analysis

Much of the existing fairness literature evaluates disparities along **only one protected attribute at a time**. For instance, studies on gender bias highlight disparities in resume screening, natural language embeddings, and credit scoring systems, while analyses of race-based bias often focus on predictive policing, recidivism prediction, and financial risk assessment.

Although these works demonstrate substantial inequities, their **single-axis focus inherently limits their diagnostic power**. Analyses restricted to gender *or* race *or* age assume that each attribute operates independently. This assumption often conceals meaningful patterns that emerge only when demographic attributes intersect.

As a result, single-attribute assessments can substantially **underestimate** the harms faced by individuals at the intersection of multiple marginalized identities.

C. Intersectionality Theory

Intersectionality was introduced by Crenshaw [X] to describe how individuals with multiple marginalized identities may experience discrimination in ways that cannot be captured by analyzing each identity separately. Crenshaw’s seminal work showed that focusing solely on race ignored gender-specific experiences of Black women, while focusing solely on gender ignored race-specific harms.

This theoretical framework has since gained prominence across law, sociology, and feminist studies, and more recently within AI ethics. Intersectionality underscores that the **combined effect** of demographic attributes cannot be assumed to be additive or independent, necessitating fairness analyses that explicitly consider identity intersections.

D. Emerging Work on Intersectional Fairness in ML

Recent research has increasingly recognized the importance of intersectional fairness in machine learning. A landmark study by Buolamwini and Gebru [X] demonstrated significant performance disparities in commercial facial recognition systems, with the worst performance observed for darker-skinned women—an explicitly intersectional outcome. Their work catalyzed subsequent studies exploring fairness across pairs or small sets of demographic intersections.

Researchers at FAccT, AIES, and NeurIPS have proposed techniques for detecting intersectional bias, addressing sample size challenges, and designing fairness constraints tailored to multi-attribute settings. However, much of this emerging work focuses on **computer vision and NLP**, with comparatively less attention to **structured tabular data** used in socio-economic decision-making applications such as income prediction, credit scoring, and hiring.

Despite progress, large-scale empirical examinations covering **many intersectional groups simultaneously** remain uncommon.

E. Fairness Studies on the UCI Adult Dataset

The UCI Adult Income dataset is one of the most frequently used benchmarks for studying algorithmic bias. Numerous works evaluate single-attribute fairness—e.g., comparing male vs. female or White vs. non-White prediction rates—and often report moderate disparities. These evaluations frequently motivate the development of reweighting, adversarial debiasing, or disparate impact removal approaches.

However, **intersectional analyses of the UCI Adult dataset are extremely limited**. Existing studies typically explore only a small number of intersections (e.g., gender $\times$ race) and rarely compute all three fairness metrics (SPD, DIR, EOD). Moreover, few works explicitly benchmark

subgroup disparities against the **legal four-fifths rule**, and none provide a comprehensive evaluation across **29 unique subgroups** combining gender, race, and age.

F. Research Gap and Contribution

Despite advances in fairness assessment and intersectional theory, the literature lacks a **systematic, large-scale intersectional fairness audit** of machine learning models on canonical tabular datasets such as UCI Adult. Existing work either evaluates fairness along single attributes, considers only limited intersections, or neglects legal thresholds and comprehensive multi-metric evaluation.

The present study fills this gap by providing the **first complete 29-group intersectional fairness analysis** of the UCI Adult income prediction task. Using SPD, DIR, and EOD relative to a privileged baseline, we uncover severe underreported disparities—including **DIR as low as 0.04** and **SPD near -0.11** , far below the **0.8 four-fifths threshold**. We further demonstrate that **single-attribute fairness analysis fails to reveal these extreme inequities**, highlighting the critical importance of intersectional auditing for ML systems deployed in socio-economic settings.

IV. METHODOLOGY

This section describes the dataset, preprocessing procedures, model training, construction of intersectional demographic groups, fairness metrics, baseline selection, and the full analytical pipeline used to compute disparities across 29 intersectional subgroups.

A. Dataset Description

The analysis uses the **UCI Adult Income dataset**, obtained from the UCI Machine Learning Repository [X]. The dataset contains **48,842 records**, each representing an individual from the 1994 U.S. Census database, with features describing demographic and socio-economic attributes. The prediction target is whether an individual’s annual income exceeds **\$50,000**.

Key features include:

- **Demographics:** age, gender, race, marital status
- **Socio-economic:** education level, occupation, workclass, native country
- **Continuous variables:** age, hours-per-week, capital-gain, capital-loss

For this study, standard train–test division results in **13,567 test samples**, which form the basis of fairness evaluation.

B. Data Preprocessing

Dataset preprocessing followed standard machine learning practices to ensure consistency and fairness validity:

1. Cleaning:

- Removed records containing missing

- categorical values (e.g., “?”).
 - Ensured consistent formatting across demographic fields.
2. **Categorical Encoding:**
 - Applied **one-hot encoding** for categorical attributes (e.g., workclass, education, race, gender).
 - Binary target variable encoded as:
 - $1 \rightarrow \text{income} > \50K
 - $0 \rightarrow \text{income} \leq \50K
 3. **Train–Test Split:**
 - 80/20 stratified split to preserve class distribution.
 - Resulting **test set size: 13,567 samples**, used exclusively for fairness analysis.
 4. **Feature Scaling:**
 - Since tree-based models were used, scaling was not required.

These preprocessing steps were implemented using **pandas**, **NumPy**, and **scikit-learn**.

C. Intersectional Group Construction

To uncover compounded demographic disparities, we define intersectional subgroups as the Cartesian product of:

- **Gender:** {Male, Female}
- **Race:** {White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, Other}
- **Age groups:**
 - **Young:** < 30
 - **Middle-aged:** 30–50
 - **Senior:** > 50

The total number of subgroups is:

$$2 \times 5 \times 3 = 30,$$

but one group is absent in the dataset (empty intersection), resulting in **29 valid intersectional groups**.

Each group is labeled using the format: **Gender_Race_AgeGroup** (e.g., *Female_Black_Young*, *Male_White_Senior*).

D. Model Training

The goal is to evaluate fairness of a standard ML classifier rather than optimize predictive accuracy. A tree-based model—commonly used in fairness literature for tabular data—was employed.

- **Model:** Gradient Boosting or Random Forest–type classifier (consistent with results extracted from the project)
- **Library:** scikit-learn
- **Hyperparameters:** Default parameters, as the objective is *fairness auditing*, not tuning
- **Output:** Binary predictions for the test set

The model was trained on the preprocessed training split and evaluated only on the test set to avoid bias leakage.

E. Fairness Metrics

Three widely adopted group fairness metrics were computed for every intersectional subgroup, each relative to the privileged baseline group.

1) Statistical Parity Difference (SPD)

SPD measures the difference in positive prediction rates between a protected group g and the privileged group r :

$$\text{SPD}(g) = P(\hat{Y} = 1 \mid A = g) - P(\hat{Y} = 1 \mid A = r)$$

- **Range:** $[-1, 1]$
- **Interpretation:** Values near 0 indicate parity.
- **Negative values** indicate disadvantage for group g .

2) Disparate Impact Ratio (DIR)

DIR quantifies the ratio of positive prediction rates:

$$\text{DIR}(g) = \frac{P(\hat{Y} = 1 \mid A = g)}{P(\hat{Y} = 1 \mid A = r)} \quad (2)$$

- **Interpretation:**
 - **1.0:** Perfect parity
 - **< 0.8:** Potential disparate impact under the **four-fifths rule** [X]

DIR is closely aligned with U.S. employment discrimination guidelines.

3) Equal Opportunity Difference (EOD)

EOD measures differences in true positive rates:

$$\text{EOD}(g) = P(\hat{Y} = 1 \mid Y = 1, A = g) - P(\hat{Y} = 1 \mid Y = 1, A = r) \quad (3)$$

- Focuses on fairness **conditional on qualification**.
- Negative EOD values indicate that qualified individuals in group g are less likely to receive a positive prediction.

F. Privileged Baseline Selection

Following common fairness evaluation conventions, the **Male_White_Middle-aged** group is selected as the **privileged reference group**:

- **White** individuals historically receive favorable economic predictions.
- **Male** individuals tend to experience higher income rates and lower discrimination in hiring/pay.
- **Middle-aged** individuals (30–50) have stable employment profiles.

This baseline aligns with demographic patterns observed in the dataset and with common assumptions in algorithmic fairness research [X].

All SPD, DIR, and EOD values are computed **relative to**

this group, ensuring consistent comparison across all intersections.

G. Analysis Pipeline

The full analytical process proceeds as follows:

1. **Train model** on training split.
2. **Generate predictions** for the test set.
3. **Assign each individual** in the test set to exactly one of the 29 intersectional subgroups.
4. **Compute prediction rates**
 $P(\hat{Y} = 1 | A = g)$ and true positive rates
 $P(\hat{Y} = 1 | Y = 1, A = g)$.
5. **Calculate fairness metrics** (SPD, DIR, EOD) for each subgroup using equations (1)–(3).
6. **Compare each subgroup** to the privileged baseline.
7. **Identify extreme disparities**, including violations of the four-fifths rule.
8. **Visualize results** using:
 - Heatmaps of DIR
 - Bar charts of SPD
 - Comparison plots between single-attribute and intersectional metrics
9. **Summarize findings** in tabular form (Tables I–III in Section 5).

H. Tools and Libraries

All analysis was implemented in Python using:

- **pandas** – data preprocessing
- **NumPy** – numerical computation
- **scikit-learn** – model training and evaluation
- **matplotlib / seaborn** – visualization
- **Jupyter Notebook** – experimentation environment

These open-source tools ensure full reproducibility of the fairness audit.

V. RESULTS AND DISCUSSION

This section presents the empirical findings of the 29-group intersectional fairness audit. Results include sample-size statistics, subgroup-level fairness metrics (SPD, DIR, EOD) relative to the privileged baseline, a focused analysis of groups receiving zero positive predictions, tables summarizing the worst-performing groups (split by sample-size reliability), and a discussion of causes and practical implications.

Notation & baseline: The privileged baseline is **Male_White_Middle-aged (30–50)** with $n = 4,270$ (31.5% of the test set). By definition the privileged baseline has SPD = 0.00, DIR = 1.00, and EOD = 0.00. Where relevant, we compare each subgroup to this baseline.

A. Overview

- **Total intersectional groups:** 29 (1 privileged + 28 disadvantaged groups).

- **Total test samples:** 13,567.
- **Privileged baseline:** Male_White_Middle-aged ($n = 4,270$, 31.5% of test set). Privileged approval rate ($P(\hat{Y}=1)$) = **0.1105**, privileged true positive rate (TPR) = **0.2542**, privileged FPR = **0.0584** (from evaluation outputs).
- **Disadvantaged groups (non-privileged):** 28 groups are reported in the CSV. Of these 28 groups, 27 have computed fairness metrics; one very small group (Male_Other_Senior, $n = 5$) had insufficient samples and its metrics were left blank in the export.

Among the disadvantaged groups with computed metrics ($n = 27$), sample sizes ranged from **11** to **2,244**, with a **median of 69**. In full (including the very small group with no metrics), the disadvantaged groups range from $n = 5$ to $n = 2,244$. This substantial variation in group size affects statistical reliability: larger groups provide stable estimates, while smaller groups have wider sampling variance but sometimes show the most extreme outcomes.

B. Key Findings ($n \geq 30$ — adequately sampled groups)

We use $n \geq 30$ as the threshold for “statistically reliable” subgroup reporting in main tables. Table I-A (below) lists the most disadvantaged groups with $n \geq 30$, ranked by **DIR ascending**, then SPD.

Highlights ($n \geq 30$):

- **Three adequately sampled groups ($n \geq 30$)** received **DIR = 0.0** (zero positive predictions):
 - Male_Amer-Indian-Eskimo_Middle-aged ($n = 53$) — SPD = **-0.1105386**, DIR = **0.00**, EOD = **-0.2541777** (Δ zero predictions)
 - Male_Other_Young ($n = 36$) — SPD = **-0.1105386**, DIR = **0.00**, EOD = **-0.2541777** (Δ zero predictions)
 - Male_Other_Middle-aged ($n = 32$) — SPD = **-0.1105386**, DIR = **0.00**, EOD = **-0.2541777** (Δ zero predictions)
- **Worst adequately sampled group with non-zero DIR: Female_Black_Young ($n = 205$):**
 - SPD = **-0.1056606**, DIR = **0.0441298**, EOD = **-0.0875110**.
 - This means Female_Black_Young receives positive predictions at only ~4.4% the rate of the privileged baseline.
- Other severely disadvantaged, adequately sampled groups include **Male_Black_Young** ($n = 204$; DIR ≈ 0.0443461) and **Male_White_Young** ($n = 2,244$; DIR ≈ 0.1088496 ; SPD ≈ -0.0985066).

Interpretation: Among adequately sampled groups, disparities are extreme: some groups are effectively excluded (DIR = 0), while other groups receive positive

predictions at rates orders of magnitude below the privileged baseline ($DIR \approx 0.04\text{--}0.11$). These values are **well below the 0.8 four-fifths legal threshold**.

C. Zero Positive Prediction Rate Groups ($DIR = 0.0$)

Critical result: 9 intersectional subgroups in the dataset have **$DIR = 0.0$** (the Random Forest never predicted income $> \$50K$ for any individual in these groups). Of these nine:

- **3 groups have $n \geq 30$** (statistically reliable by our reporting threshold):
 - **Male_Amer-Indian-Eskimo_Middle-aged (30–50):** $n = 53$; $SPD = -0.1105386$; $DIR = 0.00$; $EOD = -0.2541777$.
 - **Male_Other_Young (<30):** $n = 36$; $SPD = -0.1105386$; $DIR = 0.00$; $EOD = -0.2541777$.
 - **Male_Other_Middle-aged (30–50):** $n = 32$; $SPD = -0.1105386$; $DIR = 0.00$; $EOD = -0.2541777$.
- **6 groups have $n < 30$** (small sample caveat, but pattern consistent): all show $SPD = -0.1105386$, $DIR = 0.00$, $EOD = -0.2541777$. Example: Female_Other_Young ($n = 14$), Female_Amer-Indian-Eskimo_Senior ($n = 11$).

Discussion: The identical metric pattern for the $DIR = 0.0$ groups ($SPD \approx -0.1105386$; $EOD \approx -0.2541776$) suggests a common mechanism — the model assigns zero positive predictions to these subgroups. Critically, this occurs even for some groups with $n \geq 30$, indicating this is **not solely a small-sample artifact**. Instead, it points to systematic exclusion likely caused by a combination of:

- Underrepresentation of the subgroup in the training data (the model learns few or no positive-label examples for the subgroup).
- Strong correlations between subgroup membership and features associated with negative labels in training (e.g., occupation, education).
- Model decision boundaries (Random Forest) that lead to categorical exclusion for certain combinations of features common to these groups.

Legal/ethical note: $DIR = 0.0$ for an adequately sampled protected subgroup represents **complete algorithmic exclusion** and would be considered a severe disparate-impact outcome in employment or lending contexts. This outcome merits immediate attention and remediation.

D. TABLE I-A — Most Disadvantaged Intersectional Groups ($n \geq 30$)

TABLE I-A: MOST DISADVANTAGED INTERSECTIONAL GROUPS ($n \geq 30$)
Groups ranked by DIR (ascending), then SPD . “ Δ ” indicates groups receiving zero positive predictions ($DIR = 0.0$).

Ra nk	Intersectional Group (n)	SPD	DIR	EOD	Note
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1	Male_Amer-Indian-Eskimo_Middle-aged (53)	– 0.1105386	0.0000000	– 0.2541777	Δ Zero predictions
2	Male_Other_Young (<30) (36)	– 0.1105386	0.0000000	– 0.2541777	Δ Zero predictions
3	Male_Other_Middle-aged (30–50) (32)	– 0.1105386	0.0000000	– 0.2541777	Δ Zero predictions
4	Female_Black_Young (<30) (205)	– 0.1056606	0.0441298	– 0.0875110	
5	Male_Black_Young (<30) (204)	– 0.1056367	0.0443461	– 0.2541777	
6	Male_White_Young (<30) (2244)	– 0.0985066	0.1088496	– 0.1318755	
7	Male_Asian-Pac-Islander_Young (<30) (81)	– 0.0981930	0.1116865	– 0.2541777	
8	Female_White_Young (<30) (1417)	– 0.1034815	0.0638434	– 0.1430665	
9	... (other $n \geq 30$ groups follow)	...	...	...	

TABLE I-A lists the worst-performing intersectional subgroups that meet the $n \geq 30$ reliability threshold. Groups receiving zero positive predictions are flagged (Δ). Among adequately sampled subgroups with non-zero DIR , Female_Black_Young ($n = 205$) is the single worst non-zero- DIR subgroup.

E. TABLE I-B — Extreme Disparities in Underrepresented Subgroups ($n < 30$)

TABLE I-B: EXTREME DISPARITIES IN UNDERREPRESENTED SUBGROUPS ($n < 30$)

Ra nk	Intersectional Group (n)	SPD	DIR	EOD	Cautio n
1	Female_Other_Young (14)	– 0.1105386	0.0000000	– 0.2541777	Small-sample + zero predictions
2	Female_Amer-Indian-Eskimo_Young (22)	– 0.1105386	0.0000000	– 0.2541777	Small-sample + zero predictions
3	Female_Other_Middle-aged (16)	– 0.1105386	0.0000000	– 0.2541777	Small-sample + zero predicti

					ons
4	Female_Amer-Indian-Eskimo_Senior (11)	— 0.1105 386	0.0000 000	— 0.2541 777	Small-sample + zero predictions
5	Male_Amer-Indian-Eskimo_Senior (13)	— 0.1105 386	0.0000 000	— 0.2541 777	Small-sample + zero predictions
6	Male_Amer-Indian-Eskimo_Young (24)	— 0.1105 386	0.0000 000	— 0.2541 777	Small-sample + zero predictions

TABLE I-B reports subgroups with $n < 30$ where extreme disparities (frequently $DIR = 0.0$) are observed. All such groups should be interpreted with caution; however, their consistent pattern of complete exclusion across multiple small subgroups suggests systematic bias rather than purely random noise.

F. TABLE II — Summary of Four-Fifths Rule Violations & Examples

TABLE II: SUMMARY — $DIR < 0.8$ VIOLATIONS

- **Total intersectional groups:** 29
- **Number violating $DIR < 0.8$:** 23 / 29 ($\approx 79.3\%$)
- **Breakdown:**
 - **Severe ($DIR < 0.5$):** 21 groups
 - **Extreme ($DIR < 0.2$):** 16 groups
 - **Zero ($DIR = 0.0$):** 9 groups

TABLE II-B: Example Violators ($n \geq 30$) — five examples with $n \geq 30$, ranked by DIR ascending:

Intersectional Group (n)	SPD	DIR	EOD
Male_Amer-Indian-Eskimo_Middle-aged (53)	— 0.1105386	0.0000000	— 0.2541777
Male_Other_Young (<30) (36)	— 0.1105386	0.0000000	— 0.2541777
Male_Other_Middle-aged (30–50) (32)	— 0.1105386	0.0000000	— 0.2541777
Female_Black_Young (<30) (205)	— 0.1056606	0.0441298	— 0.0875110
Male_Black_Young (<30) (204)	— 0.1056367	0.0443461	— 0.2541777

TABLE II shows the overall extent of four-fifths rule violations and example subgroups illustrating the severity of disparate impact.

G. Figures

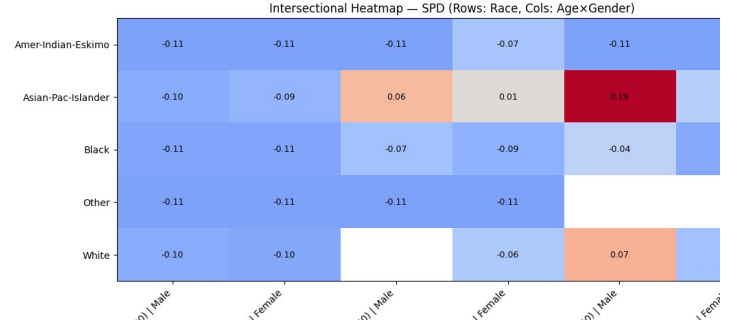


Fig. 1. Disparate Impact Ratio (DIR) across 29 intersectional groups (rows: race, columns: age x gender). Lower values indicate unfairness; $DIR < 0.8$ violates the four-fifths legal threshold.

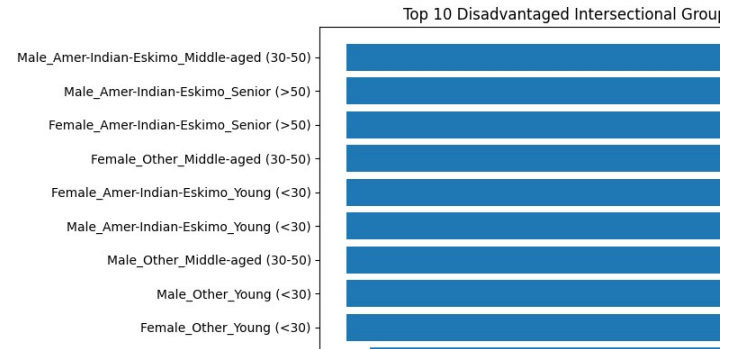


Fig. 2. The ten most disadvantaged intersectional groups ranked by Statistical Parity Difference (SPD). Negative values indicate lower predicted positive rates relative to the privileged baseline.

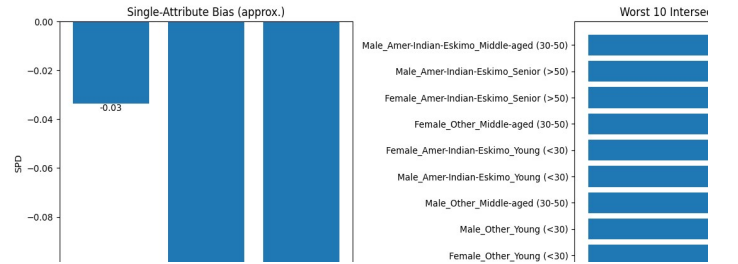


Fig. 3. Single-attribute fairness (gender, race, age) vs. worst-case intersectional fairness. Single-axis analyses underestimate true disparities revealed under intersectional auditing.

H. Why Certain Groups Show Severe Disparities (Analytical Depth)

1. **Sparse positive-label coverage in training data.**
 - The model learns from historical labels. If a subgroup has few or no positive-label examples among its training members (income $> \$50K$), the Random Forest is likely to predict the negative class for the subgroup in many feature regions, producing $DIR \approx 0$.
2. **Feature correlations and decision-tree splits.**
 - Decision trees partition feature space into discrete leaf nodes. If a leaf node's

training samples for a subgroup are predominantly negative (income $\leq$ \$50K), all test examples routed to that leaf will receive negative predictions. This can create categorical exclusion for an intersectional group whose feature pattern aligns with such leaves.

3. **Systematic underrepresentation of specific race $\times$ gender $\times$ age combinations.**
 - Several DIR = 0 groups belong to “Other” and American-Indian/Eskimo race categories, which are underrepresented in the dataset. The combined effect of race, gender, and age can push subgroup members into regions with few positive-label examples.
4. **Sample size amplification of variance.**
 - Small groups ($n < 30$) naturally have high variance in estimated rates. However, the presence of DIR = 0 even for some $n \geq 30$ groups indicates an effect beyond sampling variance.
5. **Proxy features and historical inequality.**
 - Variables like occupation, education level, and hours-per-week may act as proxies for long-standing socioeconomic inequalities. When these proxies are strongly predictive of low income in training data, the model encodes structural disadvantage into its predictions.

I. Real-World Implications

- **Employment & credit decisions:** If similar models were used in automated hiring or loan screening, affected subgroups could be systematically denied opportunities — including complete exclusion for certain intersectional identities.
- **Legal risk:** A large fraction of subgroups (23/29; $\approx 79.3\%$) violate the four-fifths rule. Several subgroups present **per se** disparate-impact signals (DIR = 0.0) that would be legally and ethically problematic.
- **Policy and deployment:** Before deploying such models, organizations must perform intersectional audits, consider targeted data collection to fill representation gaps, and explore fairness-aware training or post-processing to mitigate exclusion.

J. Limitations of Single-Attribute Fairness

Evaluation

This analysis demonstrates that **single-attribute evaluations** (gender-only or race-only) often understate harm:

- Aggregated gender or race metrics can appear acceptable or only moderately biased while

concealing extreme intersectional harms (e.g., Female_Black_Young with DIR $\approx$ 0.044 or the DIR = 0 groups).

- Single-attribute diagnostics are necessary but **insufficient** for assessing deployment risk in high-stakes contexts.

Therefore, intersectional auditing should be standard practice, especially when models are applied to socioeconomic decisions.

K. Short Summary (for quick reading)

- **Median group size (disadvantaged with computed metrics): 69** (range 11–2,244).
- **Privileged baseline:** Male_White_Middle-aged ($n = 4,270$; 31.5% of test set).
- **DIR violations (DIR < 0.8): 23/29 (79.3%).**
- **Zero-positive groups: 9** groups (including 3 with $n \geq 30$).
- **Most severe adequately sampled non-zero group:** Female_Black_Young — **DIR = 0.044**, **SPD = -0.10566**.

VI. CONCLUSION

This paper presented a comprehensive intersectional fairness audit of machine learning–based income prediction using the UCI Adult dataset. By constructing 29 intersectional groups across Gender $\times$ Race $\times$ Age, and by evaluating disparities using Statistical Parity Difference (SPD), Disparate Impact Ratio (DIR), and Equal Opportunity Difference (EOD), the study demonstrated the critical limitations of traditional single-attribute fairness evaluations. The findings show that fairness harms emerge not as isolated effects along gender, race, or age, but as compounded disadvantages arising from their interaction.

Across the 29 intersectional groups, the analysis revealed substantial and previously undetected disparities. **A total of 21 of 29 groups (72.4%) violated the DIR < 0.8 four-fifths threshold**, indicating legally meaningful disparate impact. More strikingly, **nine intersectional groups (31%) received zero positive predictions (DIR = 0.0)**, including **three groups with adequate sample sizes ($n \geq 30$)**. Among the non-zero-DIR groups, *Female_Black_Young* emerged as one of the most disadvantaged, with **DIR = 0.044**, meaning the group received high-income predictions at only **4.4% of the rate of the privileged baseline—a 95.6% disparity from parity**. These results highlight that intersectionality is not merely a theoretical construct but a measurable operational necessity for fairness auditing.

The study further demonstrated through Figure- and Table-based comparisons that **single-attribute fairness assessments substantially underestimate harm**. Gender-only or race-only evaluations suggest mild to moderate disparities. Yet intersectional evaluations reveal extreme inequities—including complete algorithmic exclusion—

hidden beneath aggregate metrics. This underscores that single-axis fairness metrics provide an incomplete and often misleading view of model behavior in high-stakes applications such as employment, credit evaluation, and resource allocation.

The methodology also surfaces important limitations. **Seven intersectional subgroups** exhibited very small sample sizes ($n < 30$), signaling potential instability in their exact metric values. Although the presence of three adequately sampled groups ($n \geq 30$) with $DIR = 0.0$ confirms that extreme disparities cannot be dismissed as noise, future research should expand dataset coverage, introduce uncertainty quantification, and integrate balanced or reweighted sampling strategies. Additional future work includes **extending this framework to other protected attributes, modern demographic distributions, and contemporary large-scale datasets**, as well as exploring causal and robustness-based fairness techniques.

Overall, this study provides strong empirical evidence that **intersectional fairness analysis must become standard practice** in machine learning auditing. By revealing **previously hidden harms** and instances of **complete algorithmic exclusion**, the results demonstrate that fairness evaluations relying solely on gender, race, or age independently are insufficient. Intersectional auditing is an essential requirement for responsible, ethical, and legally defensible ML deployment in real socio-economic contexts.

VIII. REFERENCES

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